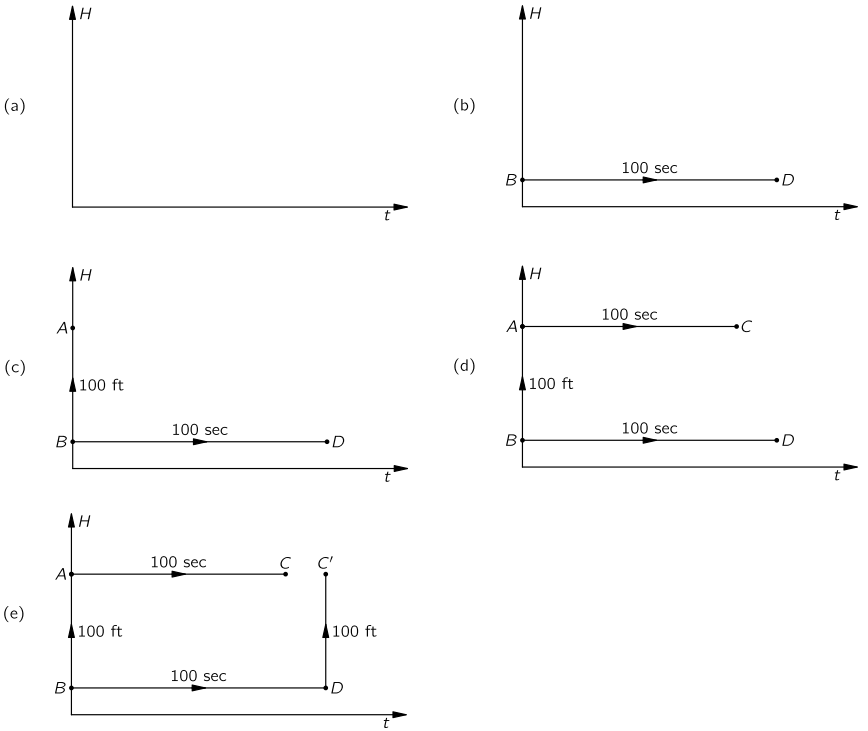

Figure 6-18 Trying to make a rectangle in space-time.

move clock *A* so that it reads the latest possible time—always assuming that it returns when *B* reads 100 seconds? You say, "That's easy. Just take *A* as high as you can. Then it will run as fast as possible, and be the latest when you return." Wrong. You forgot something—we've only got 100 seconds to go up and back. If we go very high, we have to go very fast to get there and back in 100 seconds. And you mustn't forget the effect of special relativity which causes moving clocks to *slow down* by the factor $\sqrt{1 - v^2/c^2}$. This relativity effect works in the direction of making clock *A* read *less*